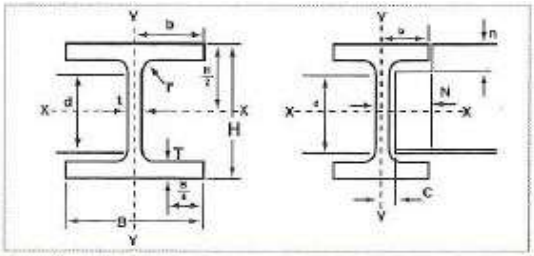
Dimensions and Properties

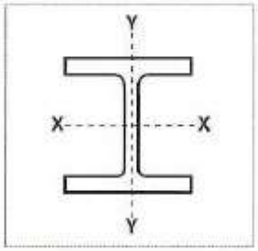


4b (i) Beams (Metric Units) - Part I

Designation		Height Of Beam	Width Of Beam	Thickness		Root Radius	Depth Between Fillets	Area Of Section	Ratios For Local Buckling		Dimensions For Detailing		
Size	Units Weight			Flange	Web				Flange	Web	End Clearance	Notch	
		H	B	T	t	r	d	A	b/T	d/t	C	N	n
mm	kg/m	mm	mm	mm	mm	mm	mm	cm^2			mm	mm	mm
100 x 50	9.30	100	50	7	5	8	70	11.8	3.57	14.0	5	33	15
125 x 60	13.2	125	60	8	6	9	91	16.8	3.75	15.2	5	37	17
150 x 75	14.0	150	75	7	5	8	120	17.8	5.36	24.0	5	45	15
150 x 100	20.7	148	100	9	6	8	114	26.3	5.56	19.0	5	57	17
	21.1	148	100	9	6	11	108	26.8	5.56	18.0	5	57	20
175 x 90	18.0	175	90	8	5	8	143	22.9	5.63	28.6	5	53	16
	18.1	175	90	8	5	9	141	23.0	5.63	28.2	5	53	17
175 x 125	23.3	169	125	8	5.5	12	129	29.7	7.81	23.5	5	70	20
200 x 100	17.8	198	99	7	4.5	8	168	22.7	7.07	37.3	4	57	15
	18.2	198	99	7	4.5	11	162	23.2	7.07	36.0	4	57	18
	20.9	200	100	8	5.5	8	168	26.7	6.25	30.5	5	57	16
	21.3	200	100	8	5.5	11	162	27.2	6.25	29.5	5	57	19
200 x 150	29.9	194	150	9	6	8	160	38.1	8.33	26.7	5	82	17
	30.6	194	150	9	6	13	150	39.0	8.33	25.0	5	82	22
	36.9	198	151	11	7	13	150	47.0	6.86	21.4	6	82	24
250 x 125	25.1	248	124	8	5	8	216	32.0	7.75	43.2	5	70	16
	25.7	248	124	8	5	12	208	32.7	7.75	41.6	5	70	20
	29.0	250	125	9	6	8	216	37.0	6.94	36.0	5	70	17
	29.6	250	125	9	6	12	208	37.7	6.94	34.7	5	70	21
250 x 175	43.6	244	175	11	7	13	196	55.5	7.95	28.0	6	94	24
	44.1	244	175	11	7	16	190	56.2	7.95	27.1	6	94	27
	51.6	248	176	13	8	16	190	65.7	6.77	23.8	6	94	29
	59.1	252	177	15	9	16	190	75.3	5.90	21.1	7	94	31
300 x 150	25.0	294	148	6	4.5	16	250	32.6	12.3	55.6	4	82	22
	32.0	298	149	8	5.5	13	256	40.8	9.31	46.5	5	82	21
	36.7	300	150	9	6.5	13	256	46.8	8.33	39.4	5	82	22
	41.4	304	150	11	6.5	13	256	52.8	6.82	39.4	5	82	24
	46.2	306	151	12	7.5	13	256	58.8	6.29	34.1	6	82	25
	69.0	318	154	18	11	13	256	87.9	4.28	23.3	8	82	31
300 x 200	48.3	290	199	10	7	18	234	61.5	9.95	33.4	6	106	28
	55.8	294	200	12	8	13	244	71.1	8.33	30.5	6	106	25
	56.8	294	200	12	8	18	234	72.4	8.33	29.3	6	106	30
	65.4	298	201	14	9	18	234	83.4	7.18	26.0	7	106	32
	77.3	304	202	17	10	18	234	98.5	5.94	23.4	7	106	35
350 x 175	41.2	346	174	9	6	13	302	52.5	9.67	50.3	5	94	22
	41.4	346	174	9	6	14	300	52.7	9.67	50.0	5	94	23
	49.4	350	175	11	7	13	302	62.9	7.95	43.1	6	94	24
	49.6	350	175	11	7	14	300	63.1	7.95	42.9	6	94	25
	57.8	354	176	13	8	14	300	73.7	6.77	37.5	6	94	27
	66.2	358	177	15	9	14	300	84.3	5.90	33.3	7	94	29
	71.8	360	178	16	10	14	300	91.4	5.56	30.0	7	94	30
	79.7	364	177	18	11	14	300	101	4.92	27.3	8	93	32
350 x 250	69.2	336	249	12	8	20	272	88.2	10.4	34.0	6	131	32
	78.1	340	250	14	9	13	286	100	8.93	31.8	7	131	27
	79.7	340	250	14	9	20	272	102	8.93	30.2	7	131	34
	94.2	346	251	17	10	20	272	120	7.38	27.2	7	131	37
	108	350	253	19	12	20	272	137	6.66	22.7	8	131	39
400 x 200	56.1	396	199	11	7	13	348	71.4	9.05	49.7	6	106	24
	56.6	396	199	11	7	16	342	72.2	9.05	48.9	6	106	27
	65.4	400	200	13	8	13	348	83.4	7.69	43.5	6	106	26
	66.0	400	200	13	8	16	342	84.1	7.69	42.8	6	106	29
	75.5	404	201	15	9	16	342	96.2	6.70	38.0	7	106	31
	88.2	410	202	18	10	16	342	112	5.61	34.2	7	106	34
	140	430	208	28	16	16	342	179	3.71	21.4	10	106	44
	187	446	214	36	22	16	342	239	2.97	15.5	13	106	52
400 x 300	94.3	386	299	14	9	22	314	120	10.7	34.9	7	155	36
	105	390	300	16	10	13	332	133	9.38	33.2	7	155	29
	107	390	300	16	10	22	314	136	9.38	31.4	7	155	38
	127	396	302	19	12	22	314	162	7.95	26.2	8	155	41
	144	402	303	22	13	22	314	184	6.89	24.2	9	155	44

Please contact us for further clarification.

Dimensions and Properties



4b (i) Beams (Metric Units) - Part II

Surface Area Per Metre	Second Moment Of Area		Radius Of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant	Designation	
	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	x-x	y-y					Unit Weight	Size
m^2	cm^4	cm^4	cm	cm	cm^3	cm^3	cm^3	cm^3	u	x	H	J	kg/m	mm
0.376	188	14.8	3.98	1.12	37.5	5.93	44.1	9.5	0.876	12.7	0.00032	2.03	9.3	**100 x 50**
0.463	413	29.2	4.95	1.32	66.1	9.75	77.6	15.7	0.872	14.0	0.00100	3.75	13.2	**125 x 60**
0.576	666	49.5	6.11	1.67	88.8	13.2	102	20.8	0.876	20.4	0.00253	2.81	14.0	**150 x 75**
0.670	1003	150	6.17	2.39	135	30.1	154	46.4	0.880	15.8	0.00727	6.56	20.7	**150 x 100**
0.665	1021	151	6.17	2.37	138	30.1	157	46.7	0.881	15.0	0.00728	7.37	21.1	
0.686	1205	97.5	7.26	2.06	138	21.7	156	33.6	0.885	21.8	0.00680	4.31	18.0	**175 x 90**
0.685	1214	97.6	7.26	2.06	139	21.7	157	33.7	0.885	21.4	0.00680	4.50	18.1	
0.806	1529	261	7.18	2.97	181	41.8	202	64.3	0.879	19.1	0.0169	6.76	23.3	**175 x 125**
0.769	1543	113	8.25	2.24	156	22.9	175	35.5	0.883	28.4	0.0103	3.30	17.8	**200 x 100**
0.764	1582	114	8.26	2.22	160	23.0	180	35.7	0.884	26.5	0.0104	3.86	18.2	
0.775	1806	134	8.23	2.24	181	26.7	205	41.6	0.878	24.9	0.0123	5.08	20.9	
0.770	1844	134	8.24	2.22	184	26.8	209	41.9	0.880	23.6	0.0124	5.77	21.3	
0.962	2625	507	8.30	3.65	271	67.6	301	103	0.876	21.2	0.0434	9.32	29.9	**200 x 150**
0.954	2690	507	8.30	3.61	277	67.7	309	104	0.878	19.8	0.0434	10.9	30.6	
0.964	3331	633	8.42	3.67	336	83.8	377	128	0.879	16.9	0.0553	18.5	36.9	
0.968	3450	255	10.4	2.82	278	41.1	312	63.2	0.884	32.0	0.0367	5.78	25.1	**250 x 125**
0.961	3537	255	10.4	2.79	285	41.1	319	63.6	0.886	29.9	0.0367	6.74	25.7	
0.974	3965	294	10.4	2.82	317	47.0	358	72.7	0.880	28.4	0.0426	8.51	29.0	
0.967	4052	294	10.4	2.79	324	47.0	366	73.1	0.881	26.9	0.0427	9.68	29.6	
1.15	6037	984	10.4	4.21	495	112	550	172	0.883	21.3	0.134	21.2	43.6	**250 x 175**
1.15	6122	985	10.4	4.19	502	113	558	173	0.884	20.5	0.134	23.2	44.1	
1.16	7308	1184	10.5	4.24	589	135	660	207	0.884	18.0	0.163	36.1	51.6	
1.17	8541	1390	10.7	4.30	678	157	764	241	0.885	16.0	0.195	53.2	59.1	
1.14	4940	326	12.3	3.16	336	44.0	375	68.4	0.878	39.7	0.0676	5.51	25.0	**300 x 150**
1.16	6318	442	12.4	3.29	424	59.4	475	91.8	0.880	35.6	0.0930	8.65	32.0	
1.16	7210	508	12.4	3.29	481	67.7	542	105	0.876	32.0	0.107	12.4	36.7	
1.17	8578	620	12.7	3.43	564	82.7	633	128	0.888	27.8	0.133	18.8	41.4	
1.18	9514	691	12.7	3.43	622	91.5	702	142	0.884	25.6	0.149	24.9	46.2	
1.21	14820	1100	13.0	3.54	932	143	1070	223	0.884	17.9	0.248	79.0	69.0	
1.33	9431	1317	12.4	4.63	650	132	721	203	0.882	26.0	0.258	22.8	48.3	**300 x 200**
1.35	11110	1602	12.5	4.75	756	160	842	245	0.882	24.0	0.319	31.4	55.8	
1.34	11340	1604	12.5	4.71	771	160	859	247	0.883	22.7	0.319	35.8	56.8	
1.35	13310	1900	12.6	4.77	893	189	1000	291	0.884	20.1	0.383	53.4	65.4	
1.37	16280	2341	12.9	4.88	1071	232	1204	356	0.888	17.2	0.482	87.8	77.3	
1.35	11040	792	14.5	3.88	638	91.0	712	140	0.881	38.1	0.225	13.2	41.2	**350 x 175**
1.35	11100	792	14.5	3.88	641	91.0	716	140	0.882	37.5	0.225	13.6	41.4	
1.36	13500	984	14.6	3.96	771	113	864	173	0.883	32.2	0.283	22.4	49.4	
1.36	13560	985	14.7	3.95	775	113	868	174	0.884	31.8	0.283	23.0	49.6	
1.37	16100	1184	14.8	4.01	909	135	1022	208	0.885	27.6	0.344	36.1	57.8	
1.38	18710	1390	14.9	4.06	1045	157	1180	243	0.886	24.4	0.409	53.6	66.2	
1.39	20240	1508	14.9	4.06	1124	169	1276	263	0.883	22.9	0.446	66.3	71.8	
1.39	22760	1669	15.0	4.06	1250	189	1425	293	0.884	20.6	0.499	91.3	79.7	
1.62	18510	3093	14.5	5.92	1102	248	1215	380	0.881	25.8	0.812	44.6	69.2	**350 x 250**
1.64	21230	3649	14.6	6.05	1249	292	1382	445	0.881	24.2	0.969	58.0	78.1	
1.63	21680	3652	14.6	6.00	1275	292	1412	447	0.882	22.8	0.970	66.3	79.7	
1.64	26440	4488	14.8	6.12	1528	358	1699	547	0.886	19.5	1.21	109	94.2	
1.65	30190	5138	14.8	6.12	1725	406	1935	623	0.882	17.6	1.41	155	108	
1.55	19770	1447	16.6	4.50	999	145	1114	223	0.883	36.8	0.536	25.0	56.1	**400 x 200**
1.55	20020	1448	16.7	4.48	1011	145	1128	224	0.884	35.6	0.536	27.1	56.6	
1.56	23460	1736	16.8	4.56	1173	174	1313	267	0.885	31.8	0.650	39.6	65.4	
1.56	23710	1737	16.8	4.54	1185	174	1326	268	0.886	30.9	0.650	42.2	66.0	
1.57	27490	2035	16.9	4.60	1361	202	1528	312	0.887	27.4	0.770	62.3	75.5	
1.58	33060	2478	17.2	4.70	1612	245	1815	378	0.891	23.4	0.952	101	88.2	
1.63	54850	4216	17.5	4.86	2551	405	2941	632	0.887	15.7	1.70	373	140	
1.68	75250	5919	17.8	4.98	3374	553	3968	873	0.883	12.5	2.49	825	187	
1.91	33680	6246	16.7	7.21	1745	418	1918	637	0.880	25.8	2.16	79.9	94.3	**400 x 300**
1.94	37860	7204	16.9	7.35	1942	480	2141	730	0.879	24.5	2.52	100	105	
1.92	38680	7210	16.9	7.28	1983	481	2188	733	0.881	23.2	2.52	114	107	
1.94	46660	8735	17.0	7.35	2357	578	2620	884	0.880	20.0	3.10	185	127	
1.95	54420	10220	17.2	7.45	2707	674	3022	1030	0.883	17.7	3.69	272	144	